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ORIGINAL PAPER

Optimization of a green bipropellant with monoethanolamine, 2-propanol, and hydrogen peroxide 90 wt% as oxidizer

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Abstract

Environmental and safety concerns have driven the search for green alternatives to conventional hypergolic propellants, particularly hydrazine and its derivatives. This study presents the development and optimization of a green bipropellant composed of monoethanolamine (MEA), 2-propanol, and copper(II) nitrate trihydrate, using hydrogen peroxide (H_2O_2) 90 wt% as the oxidizer. A Rotatable Central Composite Design (RCCD) was applied to minimize the ignition delay time (IDT), resulting in an optimized formulation with a minimum IDT of 22 ms. The inclusion of 2-propanol enhanced ignition behavior and combustion performance. The optimized formulation exhibited a specific impulse of 168 s, a characteristic velocity of 1523 m/s, a density-specific impulse ($\rho_b I_{sp}$) of 216 s g/cm^3, a flash point of 27.5 °C, and a calorific value of 23170 J/g. UV-Vis spectroscopy confirmed the formation of the $[Cu(MEA)_4]^{2+}$ complex, which catalyzes the decomposition of H_2O_2 and enables hypergolic ignition. These results underscore the potential of MEA/2-propanol-based formulations as efficient, safer, and environmentally friendly alternatives to conventional hypergolic bipropellants, particularly for applications that demand reduced toxicity, improved ignition characteristics, and enhanced volumetric performance.

Keywords Green propellant · Bipropellant · Design of experiments · Hydrogen peroxide · CEA-NASA · Liquid propellant

Igor Mendes de Aguiar Lysakovski Bampi, Victor Rabelo Rodrigues, Leonardo Henrique Gouvea and Luís Gustavo Ferroni Pereira are Investigation, Methodology, Supervision, and Review.

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Introduction

Currently, chemical propulsion is the primary means of accessing space due to its advantages over electric propulsion systems. Chemical propulsion converts thermal energy from chemical reactions into the kinetic energy of combustion products and can be classified into solid, liquid, and hybrid propellants (DeLuca et al. 2017). Liquid propulsion systems offer advantages such as high specific impulse and throttle capability, making them widely used in large launch vehicles (Heister et al. 2019). Liquid propellants are primarily used in rocket engines and are classified as monopropellants or bipropellants. Monopropellants use a single fluid that decomposes over a catalyst, providing simpler operation but limited performance. They are commonly employed in satellite and spacecraft attitude control due to their compact and reliable design (Sutton and Biblarz 2001; de Iaco Veris 2021). Bipropellant systems, which store fuel and oxidizer separately and mix them in the combustion chamber, offer high performance and are used in most large rockets (Sutton and Biblarz 2001; Heister et al. 2019). These systems

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are preferred when missions demand high total impulse (Heister et al. 2019). Some bipropellant combinations are hypergolic, meaning that the fuel and oxidizer ignite spontaneously upon contact under ambient conditions (Davis and Yilmaz 2014). This feature is particularly beneficial for upper-stage engines, as it allows multiple restarts, increases trajectory precision, and reduces ignition system complexity (Kamal et al. 2011). As a result, hypergolic systems lower both failure probability and payload mass (Davis and Yilmaz 2014), in addition to simplifying ignition sequences and reducing costs (Silva and lha 2012).

Among hypergolic systems, ignition delay time (IDT), the time between the initial contact of fuel and oxidizer and the start of ignition, is one of the most critical parameters (Davis and Yilmaz 2014). An excessively long IDT can result in combustion outside the chamber or hard starts. Therefore, the IDT must be optimized to ensure safe and efficient engine operation (Farmer et al. 2004; Slocum-Wang et al. 2006). For liquid rocket engines, the IDT should generally be less than 100 ms to guarantee proper combustion chamber performance (Wang et al. 2020; Florczuk and Rarata 2017). The choice of bipropellant is influenced not only by performance and efficiency, but also by handling, toxicity, cost, availability, and storability (Huzel and Huang 1971).

Parameters that are crucial in propellant formulation include specific impulse (I_{sp}), propellant bulk density (fuel mixture) (ρ_b), density-specific impulse ($\rho_b Isp$), and characteristic velocity (c^*), as key indicators of propellant performance and efficiency. I_{sp}, defined as the thrust delivered per unit weight flow of propellant consumed (Sutton and Biblarz 2001), directly relates to the engine's thrust capability. ρ_b is written in terms of fuel and oxidizer ratios, as well as the mixture ratio. This parameter can determine the total volume of tankage required. High bulk density is desirable, as a smaller volume of tankage will be required to store a given mass (Heister et al. 2019). Density-specific impulse ($\rho_b I_{sp}$), the product of propellant bulk density (ρ_b) and I_{sp}, allows for performance comparison based on impulse per unit volume, with high-density propellants contributing to reduced tank mass requirements (Sutton and Biblarz 2001). $\rho_b I_{sp}$ provides a density-normalized efficiency metric, while c^*, typically ranging from 1500 to 2500 m/s, reflects combustion efficiency (Sforza 2017). The oxidizer-to-fuel mass ratio (O/F) also plays a central role in determining optimal performance and efficiency parameters, such as I_{sp} and c^*.

Typical hypergolic fuels include hydrazine (N_2H_4), monomethylhydrazine (MMH), unsymmetrical dimethylhydrazine (UDMH), ethanol, methanol, kerosene, liquid hydrogen, and liquefied methane. These are often combined with oxidizers such as white fuming nitric acid (WFNA), red fuming nitric acid (RFNA), liquid oxygen, nitrogen tetroxide (NTO), or Mixed Oxides of Nitrogen (MON) (Heister et al. 2019; Kamal et al. 2011; Isaac Sam et al. 2022; Wright and CO 1977). However, many of these traditional propellants pose serious health and environmental risks. Hydrazine and its derivatives are highly toxic and carcinogenic, requiring costly handling procedures, while NTO and nitric acid are corrosive and hazardous (Turner 2005; Kamal et al. 2011).

Hydrogen peroxide (H_2O_2) at concentrations above 87%, known as high-test peroxide (HTP), has been used as an oxidizer since 1934 due to its favorable density, reactivity, and safety profile (Negri and Lauck 2022). HTP is non-toxic, non-carcinogenic, cost-effective, and logistically manageable (Gotzig 2017). Driven by environmental and safety concerns, many studies have explored non-toxic, high-performance alternatives to hydrazine. These efforts aim to develop propellants with low volatility, good thermal stability, high density, efficient combustion, and short IDT (Sarritzu et al. 2024; Caffiero et al. 2024; Wang et al. 2022; Zhao et al. 2021). In addition, hydrogen peroxide can also function as an oxidizer in monopropellant (Elbasuney et al. 2024) and bipropellant systems (Sutton and Biblarz 2001). HTP has a higher density compared to hydrazine, which contributes to its appeal as a propellant. Its safer handling and transport characteristics also help reduce operational costs significantly. At 90 wt% concentration, hydrogen peroxide achieves a density-specific impulse of approximately 232 s, which is comparable to that of hydrazine (230 s) (Palmer 2014; Bonifacio et al. 2014). In environmentally friendly bipropellant configurations, it can function effectively as an oxidizer when paired with hydrocarbon fuels (Ricker et al. 2019). However, one of its limitations is the absence of spontaneous ignition when combined with such fuels, making hypergolicity a challenge (Elbasuney et al. 2024).

Elbasuney et al. (2024) investigated this limitation by developing environmentally friendly nano-catalysts to improve the decomposition of hydrogen peroxide in green propellant systems. Silver (Ag) and manganese dioxide (MnO_2) nanoparticles were synthesized using sustainable methods and characterized by UV-Vis, TEM, SEM, XRD, and EDAX. While Ag exhibited rapid ignition performance, it degraded quickly due to oxidation and poisoning. In contrast, MnO_2 demonstrated slower activation but significantly greater durability, rendering it more suitable for reusable catalytic beds in monopropellant or bipropellant configurations.

Zhao et al. (2021) developed and characterized halogen-free energetic complexes that show excellent thermal stability and hypergolic behavior, making them promising candidates for green bipropellants. Wang et al. (2022) proposed a design strategy for hypergolic promoters based on cation-anion synergy, using imidazolium-substituted borohydride cations and iodocuprate(I) anions for rapid ignition with 90 wt% H_2O_2. Caffiero et al. (2024) investigated the ignition behavior and performance of MEA-based green bipropellants, exploring copper-based catalysts, peroxide concentrations, and ethanol as a co-solvent. Their findings showed trade-offs between gravimetric and volumetric performance and emphasized the importance of additive solubility and alcohol composition. Sarritzu et al. (2024) developed HIP11, a low-toxicity hypergolic ionic propellant, based on EMIM SCN and CuSCN, which demonstrated rapid ignition with hydrogen peroxide and promising safety properties.

Monoethanolamine (MEA) stands out as a promising candidate for green propellants due to its significantly lower toxicity when compared to monomethylhydrazine (MMH) and unsymmetrical dimethylhydrazine (UDMH). This lower toxicity is evidenced by its higher values of lethal dose (LD_{50}) and lethal concentration (LC_{50}), which are common indicators used in toxicological assessments. LD_{50} refers to the single dose of a substance that causes the death of 50% of a group of test animals and serves as a standard metric for acute toxicity. LC_{50}, on the other hand, denotes the concentration, usually in air, although occasionally in water for environmental studies, that proves fatal to half of a test population within a defined exposure period, usually four hours. After exposure, animals are monitored for up to 14 days to observe any delayed toxic effects (Canadian Centre for Occupational Health and Safety 2021). Comparative studies confirm that MEA's higher LD_{50} (1515 mg/kg, oral, rat) and LC_{50} (1215 ppm, 4h, inhalation, rat) values reflect substantially lower health risks relative to MMH (LD_{50} = 25 mg/kg, oral, rat; LC_{50} = 98 ppm, 4h, inhalation, rat) and UDMH (LD_{50} = 62 mg/kg, oral, rat; LC_{50} = 360 ppm, 4h, inhalation, rat) (National Center for Biotechnology Information 2024a, b, c).

When combined with H_2O_2, MEA shows rapid ignition and acceptable efficiency and performance (Melof and Grubelich 2001; Ak et al. 2011; Caffiero et al. 2024). Melof and Grubelich (2001) reported MEA's freezing point at 10.3 °C, while 90% H_2O_2 freezes at -11.5 °C. This challenge can be addressed by adding alcohols like ethanol, methanol, furfuryl alcohol, or 2-propanol, which improve fluidity and ignition behavior.

The use of 2-propanol as an additive in hypergolic fuel formulations presents several advantages. As a secondary alcohol, 2-propanol exhibits higher combustion heat (–2004 kJ/mol) compared to primary alcohols such as ethanol (–1367 kJ/mol), which can contribute to improved energetic efficiency (National Institute of Standards and Technology (NIST) 2024a, b). It also has lower hygroscopicity and volatility, improving formulation stability during storage and handling, an important factor in operational reliability [35,36]. Furthermore, 2-propanol is miscible with monoethanolamine (MEA) and is compatible with metal-based catalysts such as copper (II) salts, facilitating homogeneous mixtures and potentially more consistent ignition behavior (Maschio et al. 2018; Mota et al. 2022). Although it exhibits a slightly higher boiling point than ethanol, the thermal energy released during the decomposition of high-test peroxide (HTP), a highly concentrated form of hydrogen peroxide, may be sufficient to promote rapid vaporization and subsequent reaction. Despite these promising characteristics, the use of 2-propanol in green hypergolic bipropellant systems remains underexplored, representing a valuable opportunity for further investigation."

This study aims to develop and optimize a novel hypergolic formulation based on monoethanolamine (MEA) and 2-propanol, using copper(II) nitrate trihydrate ($Cu(NO_3)_2 \cdot 3H_2O$) as a catalyst and 90 wt% H_2O_2 as the oxidizer, as a safer and more efficient alternative to conventional toxic bipropellants. The formulation was optimized using a Rotatable Central Composite Design (RCCD) to enhance the specific impulse (I_{sp}) and characteristic velocity (c^*) through performance modeling with NASA CEA, and subsequently characterized by thermophysical and spectroscopic analyses.

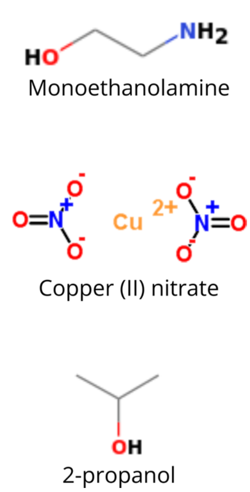

Fig. 1 Chemical structure of the reagents

Methods

Chemicals and reagents

Monoethanolamine (MEA, 99% purity), cas number 141-43-5, was used as the base fuel and was purchased from Vetec Química Fina Ltda. Copper(II) nitrate trihydrate ($Cu(NO_3)_2 \cdot 3H_2O$, 98%), cas number 10031-43-3, was used as the catalyst and was obtained from Labsynth Produtos para Laboratórios Ltda. 2-Propanol (99.5%), cas number 67-63-0, was employed as a co-fuel and sourced from Dinâmica Química. Fig.1 shows the chemical structure of the reagents.

Hydrogen peroxide (H_2O_2, Interox 60) was supplied by Solvay and concentrated to 90 wt% using distillator equipment available at the Propellants Research Laboratory (LAPEP) of the Aeronautics Institute of Technology (ITA). The methodology used was based on the approaches described in Greene et al. (2005); Pelin et al. (2020); Rarata et al. (2016). The final concentration was measured using a DMA 35 density meter from Anton Paar, and the calculation method was taken from Schumb et al. (1955).

Liquid propellant preparation

To prepare the fuel, safety personal equipment were necessary, such as lab coat, goggles, gloves, and the fume hood to handle the volatile reagents (MEA and 2-propanol). First, copper (II) nitrate trihydrate was weighed on an analytical balance using weighing paper and transferred into a beaker containing MEA previously weighed on a semi-analytical balance. Then, both reagents were mixed in a beaker using a magnetic stirrer for around 30 minutes or until the salt was completely dissolved. After complete dissolution, 2-propanol was weighed on a semi-analytical balance into the beaker containing MEA and copper (II) nitrate trihydrate and stirred for around 10 minutes or until a homogeneous mixture was obtained.

Drop test

The drop test was used to determine the hypergolic behavior of the liquid fuel when in contact with the oxidizer. A formulation is considered hypergolic if it ignites spontaneously upon contact with hydrogen peroxide. To evaluate this property, the ignition delay time (IDT) was measured for each formulation.

All personnel wore appropriate personal protective equipment, including safety goggles, gloves, and lab coats. A safety perimeter was established around the drop zone, and all experiments were performed under controlled conditions inside a fume hood to ensure proper ventilation and to minimize exposure to vapors. The equipment was positioned at a fixed distance from the reaction vessel, and a high-speed camera recorded the moment of contact and ignition. The burette was positioned 15 cm above the fuel to ensure that the flame produced during ignition would not reach the oxidizer contained in the burette, thereby maintaining safe operating conditions. The reaction vessel was placed on a non-flammable surface, and fire-suppression equipment was kept readily available as a precaution.

For each test, 5 mL of the liquid fuel was transferred to a beaker. A volume of 5 mL was selected because preliminary trials showed that this amount provided higher reproducibility compared with smaller volumes. A single drop (0.05 mL) of 90 wt% H_2O_2 was released from a burette positioned 15 cm above the fuel. The test was recorded using a high-speed camera operating at 1000 frames per second (fps). The drop velocity on impact was 1.7 m/s. The error of the IDT determination method is $\pm$ 1 ms. The schematic representation of the drop test setup is shown in Fig. 2.

Figure 3 illustrates the experimental setup and ignition sequence observed during the hypergolic drop test. The sequence begins with the release of a drop of hydrogen peroxide oxidizer (a), which falls and makes contact with the liquid fuel layer (b). Upon contact, initial vapor release is observed due to the catalytic decomposition of H_2O_2 (c), followed by a progressive increase in vapor generation (d). As the reaction intensifies, reactive clusters form at the interface (e), marking the onset of an exothermic reaction. This leads to visible ignition (f), rapid flame propagation through the reaction zone (g), and finally the establishment of a fully developed and stable flame (h).

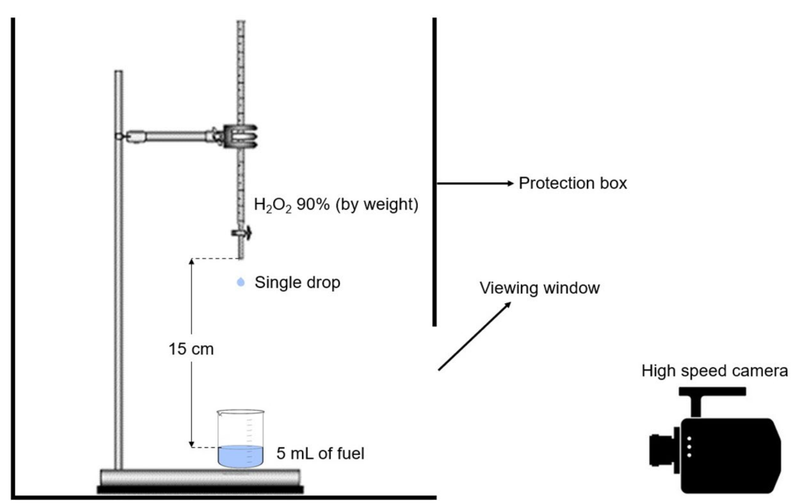

Fig. 2 Drop test scheme

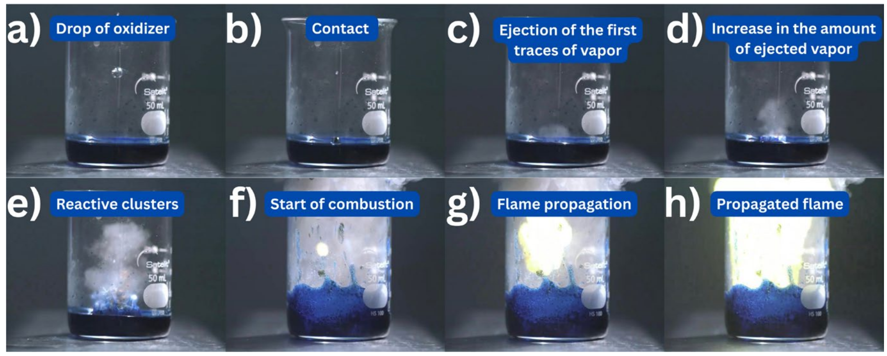

Fig. 3 **a**) Drop of oxidizer (hydrogen peroxide) falling; **b**) Contact with the fuel; **c**) Initial vapor release from H_2O_2 decomposition; **d**) Increased vapor formation; **e**) Reactive cluster formation (exothermic reaction); **f**) Onset of ignition; **g**) Flame propagation; **h**) Fully developed flame

Experimental region delimitation

To ensure all tested compositions exhibited homogeneous hypergolic behavior without catalyst precipitation, the upper and lower bounds of MEA, copper (II) nitrate trihydrate, and 2-propanol concentrations were determined. Maschio et al. (2018) demonstrated that mixtures of MEA and copper(II) nitrate trihydrate are hypergolic (Maschio 2017; Maschio et al. 2018). Based on this, the lower limit for 2-propanol was set at 0 %.

The lower and upper concentration limits of copper(II) nitrate trihydrate were determined by preparing binary mixtures with MEA. The lower limit corresponds to the minimum concentration of the catalyst in MEA capable of initiating hypergolic ignition (Point A), whereas the upper limit refers to the maximum concentration at which it remains soluble while still exhibiting hypergolic behavior (Point B).

The upper limit of 2-propanol was established by fixing copper (II) nitrate trihydrate at its lower limit and gradually replacing MEA with 2-propanol until hypergolic behavior ceased. The highest concentration at which ignition still occurred was defined as the upper limit of 2-propanol (Point C).

Since the solubility of copper (II) nitrate trihydrate depends on the MEA/2-propanol ratio, the alcohol was fixed at its upper limit, and increasing amounts of the salt were added until precipitation was observed. This procedure enabled the identification of the lower concentration limit for MEA (Point D). Together, the four boundary points (A, B, C, and D) defined the experimental region, and each composition within this region was validated through drop test experiments.

Rotatable central composite design (RCCD)

A Rotatable Central Composite Design (RCCD) was employed to optimize the ignition delay time (IDT) in the bipropellant system containing copper (II) nitrate trihydrate, 2-propanol, and MEA. Two independent variables (factors) were considered: the concentration (wt%) of copper (II) nitrate trihydrate (Cu^{2+}) (x_1), and the concentration (wt%) of 2-propanol (x_2). MEA was used to complete the mixture to a total of 100%. The RCCD consisted of a full 2^2 factorial design, four axial points (±1.41), and three center points, resulting in 11 total experiments (Fig. 4). In RCCD, replicates are typically performed at the center point of the experimental domain rather than at all design points. This strategy offers several advantages. First, center point replicates provide an independent estimate of experimental error (pure error), which is essential for assessing the adequacy of the fitted model and for conducting lack-of-fit tests. Second, replication at the center improves the precision of predictions near the center of the design space, where the optimum response is often located (Montgomery 2008). According to Montgomery (Montgomery 2008), including multiple runs at the center point allows for effective detection of curvature in the response surface and strengthens the statistical validity of the design without significantly increasing the number of required experimental runs.

The experimental design matrix (coded levels) used in the RCCD are shown in Table 1. Axial levels (±1.41) were

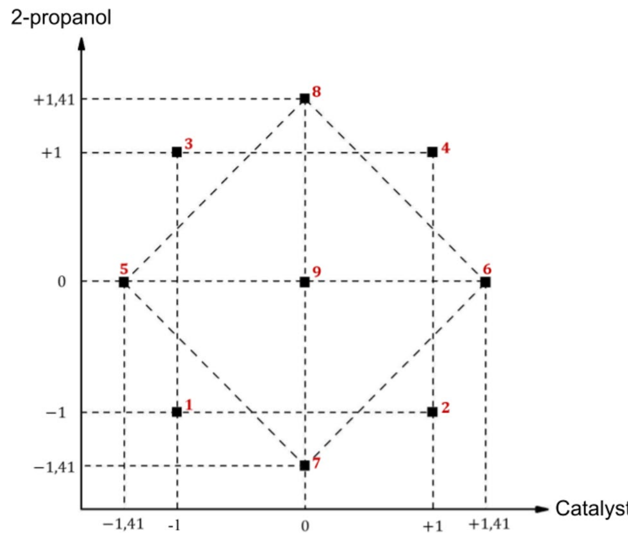

Fig. 4 Schematic representation of RCCD

Table 1 Experimental design matrix (coded levels) used in the RCCD

Run	X_1 ($Cu(NO_3)_2 \cdot 3H_2O$)	X_2 (2-Propanol)
1	-1	-1
2	+1	-1
3	-1	+1
4	+1	+1
5	-1.41	0
6	+1.41	0
7	0	-1.41
8	0	+1.41
9	0	0
10	0	0
11	0	0

set according to the experimental region boundaries, while other levels followed RCCD structure.

In order to ensure that the observations or errors are independently distributed random variables, all experiments were conducted in a random order. Conducting the experiments in a random sequence supports the validity of the assumption that errors are independent. Furthermore, this strategy helps to reduce the impact of uncontrolled external factors by distributing their effects across the trials (Montgomery 2008).

A second-order polynomial regression model was used to fit the IDT response, as shown in Eq. 1:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 \quad (1)$$

Where y is the IDT, x_1 and x_2 are the coded independent variables, β_0 is the intercept, β_1 and β_2 are linear coefficients, β_{11} and β_{22} are quadratic coefficients, and β_{12} is the interaction coefficient.

The statistical significance of the model was evaluated using Analysis of Variance (ANOVA), which decomposes the total variability in the data into components attributed to the model and to experimental error. This method compares the variance explained by the model to the residual variance using the F-ratio as a statistical criterion. When the F-value exceeds a critical threshold determined by the degrees of freedom and the chosen significance level, it indicates that the effects of the factors are statistically significant relative to the experimental noise. Thus, ANOVA helps determine whether the model reliably captures the influence of independent variables on the response, ensuring the adequacy of the regression fit (Montgomery 2008). Response surface and contour plots were generated to identify the conditions that maximize I_{sp} and c^*, and to minimize the ignition delay time (IDT).

Fuel characterization

CEA-NASA simulation

The performance and efficiency of the optimized formulation was evaluated using NASA's Chemical Equilibrium with Applications (CEA) software [45]. This program calculates product concentrations at chemical equilibrium and provides thermodynamic and transport properties for the resulting mixture.

The rocket performance was evaluated under equilibrium conditions at a chamber pressure of 5 bar and an exit pressure of 1.02 bar, resulting in a pressure ratio of 4.9 (Maschio et al. 2020). Simulations were carried out in various mass ratios of oxidizer to fuel using the optimized RCCD formulation, with an initial temperature of 298 K, depending on the physical state of the components: where L denotes liquid and s denotes solid. Table 2 presents the standard enthalpy of formation for each species. As the values for copper(II) nitrate trihydrate and MEA are not available in the NASA CEA database, data from the literature were used instead. For the remaining species, values were obtained from the NASA CEA library.

Catalytic complex identification

The UV-Vis absorption spectrum of the fuel was recorded using a Bel Photonics UV-51 spectrophotometer. For this analysis, a representative fuel sample containing monoethanolamine (MEA), copper(II) nitrate trihydrate, and 2-propanol was prepared under optimized conditions.

This procedure aimed to investigate the coordination environment of copper (II) ions in the presence of MEA, which acts as both a fuel and a ligand. Under alkaline and ligand-rich conditions, copper (II) ions are known to

Table 2 Standard Enthalpy of Formation (kJ/mol) at T = 298 K for the species used in NASA CEA

Species Name	Formula	MW (g/mol)	ΔH_f° (kJ/mol)	Refs.
MEA (L)	C_2H_7NO	61.08	-507.50	NIST Chemistry WebBook (2024)
Cu^{2+} (S)	$Cu(NO_3)_2 \cdot 3H_2O$	241.6	-1217	Varma et al. (2016)
2-Propanol (L)	C_3H_8O	60.10	-272.70	McBride et al. (2002)
Ethanol (L)	C_2H_6O	46.07	-277.51	McBride et al. (2002)
Hydrogen peroxide (L)	H_2O_2	34.01	-187.78	McBride et al. (2002)
Water (L)	H_2O	18.02	-286.30	McBride et al. (2002)

form various coordination complexes with ethanolamines, including $[Cu(MEA)]^{2+}$, $[Cu(MEA)_2]^{2+}$, $[Cu(MEA)_3]^{2+}$, and $[Cu(MEA)_4]^{2+}$ (Casassas et al. 1989; Maschio 2017; Maschio et al. 2018). These complexes are the result of the addition of stepwise ligands and may involve deprotonation and chelation mechanisms (Casassas et al. 1989).

The identification of such complexes is relevant for understanding the catalytic role of copper (II) in hydrogen peroxide decomposition. The exothermic decomposition of H_2O_2 catalyzed by Cu(II)-MEA complexes produces oxygen, water vapor, and heat, which can facilitate fuel vaporization and promote hypergolic ignition (Bonifacio et al. 2014). Thus, UV-Vis spectroscopy was employed to support the structural characterization of these complexes under formulation-relevant conditions.

Density, calorific power, and flash point

To compare the physical characteristics of formulations with and without 2-propanol, two optimized fuels were prepared. Density, calorific power, and flash point were measured. The characterization of liquid propellants requires evaluating key physicochemical properties that influence performance, safety, and operational feasibility. Density (also referred to as specific mass) directly impacts the volumetric efficiency of the propellant, playing a central role in determining tank size and contributing to the calculation of $\rho_b I_{sp}$ (Heister et al. 2019). Calorific power (measured in J/g) reflects the amount of energy released during combustion per unit mass of fuel and is a critical parameter for assessing the theoretical performance and thermal behavior of the formulation (Turns 2012). Flash point, defined as the lowest temperature at which a liquid generates sufficient vapor to ignite in air, is a primary safety indicator (Association 2018). Lower flash points are generally associated with increased flammability, influencing handling, storage, and ignition characteristics. Together, these parameters provide a comprehensive understanding of the propellant's energetic potential, density-normalized performance, and ignition behavior under ambient conditions.

To experimentally assess these parameters, the following procedures were conducted for both formulations:

Density was measured using a Rudolph DDM 2911 density meter at 20 °C. Approximately 2 mL of each sample was injected into the equipment.

Calorific power was determined using an IKA®C1 compact bomb calorimeter. About 0.4 g of each formulation was combusted in an oxygen-rich environment, and energy release was measured by water temperature change.

Flash point was measured with a Normalab NPM 450 closed-cup tester, following the device's standard protocol. Around 70 mL of sample was heated at a rate of 1 °C/min under continuous stirring. An electric spark was periodically introduced until visible ignition occurred; this temperature was recorded as the flash point.

Theoretical density of the fuel mixture (ρ_b) and density-specific impulse ($\rho_b I_{sp}$)

The bulk density of a propellant (ρ_b) is a critical parameter that quantifies the mass of propellant per unit volume (Heister et al. 2019). It can be determined based on the densities of the individual components and the oxidizer-to-fuel mixture ratio (*O*/*F*), as expressed in Equation 2:

$$\rho_b = \frac{1 + O/F}{\frac{1}{\rho_f} + \frac{O/F}{\rho_{ox}}} \tag{2}$$

In this equation, *O*/*F* denotes the oxidizer-to-fuel mass ratio, which represents the proportion between the mass of oxidizer and the mass of fuel in the bipropellant mixture. The term ρ_f refers to the density of the fuel (g/cm^3), corresponding to its liquid-phase density at a standard temperature, typically 20 °C. Similarly, ρ_{ox} is the density of the oxidizer (g/cm^3), such as hydrogen peroxide, also measured at the same reference temperature.

This property is fundamental for estimating the volumetric requirements of storage and feed systems. A higher bulk density is generally advantageous, as it allows for a reduction in tank volume for a given mass of propellant, contributing to more compact and lightweight propulsion system designs (Heister et al. 2019).

The volumetric specific impulse, expressed as $\rho_b I_{sp}$, is defined as the product of the propellant mixture density (ρ_b) and the specific impulse (I_{sp}):

$$\rho_b I_{sp} = \rho_b \cdot I_{sp} \tag{3}$$

While I_{sp} provides a measure of propulsion efficiency per unit mass, $\rho_b I_{sp}$ quantifies the impulse delivered per unit volume of propellant. This parameter is particularly relevant in applications where tank volume is constrained, such as upper-stage engines, satellite propulsion systems, or compact launch vehicles. A higher $\rho_b I_{sp}$ indicates improved volumetric efficiency, enabling greater thrust delivery without increasing the structural volume required for propellant storage. As noted by Heister et al. (2019), this metric is critical for optimizing performance in systems where mass and volume minimization are both essential design considerations.

Table 3 Analyzed formulations and their respective proportions (in wt%)

	A	B	C	D
Monoethanolamine, wt%	96	87	50	45
Copper(II) nitrate, wt%	4	13	4	9
2-Propanol, wt%	0	0	46	46

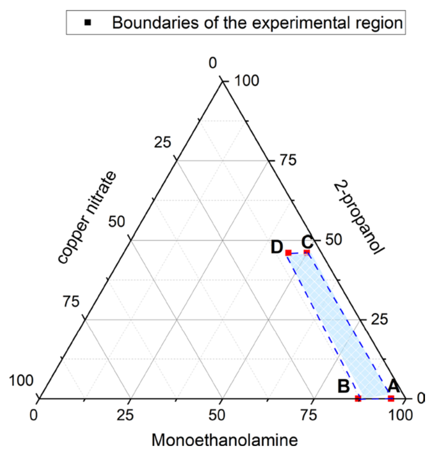

Fig. 5 Ternary diagram representing the mass proportions of the experimental region

Table 4 IDT measured in RCCD trials

Experiment	Independent variables (Factors)				Response IDT (ms)
	Uncoded values		Coded values		
	Copper(II) nitrate	2-Propanol	x_1	x_2	
1 (3)	5.3	6.7	-1	-1	45 ± 1
2 (7)	11.7	6.7	+1	-1	37 ± 1
3 (11)	5.3	39.3	-1	+1	28 ± 1
4 (4)	11.7	39.3	+1	+1	25 ± 1
5 (1)	4.0	23.0	-1.41	0	52 ± 1
6 (5)	13.0	23.0	+1.41	0	39 ± 1
7 (10)	8.5	0.0	0	-1.41	45 ± 1
8 (8)	8.5	46.0	0	+1.41	28 ± 1
9 (2)	8.5	23.0	0	0	24 ± 1
10 (9)	8.5	23.0	0	0	22 ± 1
11 (6)	8.5	23.0	0	0	28 ± 1

Results and discussion

Experimental region delimitation

Table 3 shows the proportions of the progressive solutions that exhibited hypergolic behavior. According to the principles of Design of Experiments (DoE), all combinations within the design space must be assessed (Montgomery 2008). Therefore, the minimum and maximum amounts of each compound were tested to determine the concentration limits necessary to ensure hypergolicity.

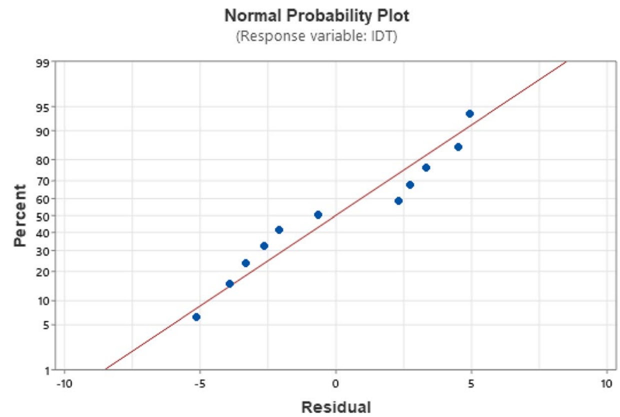

Fig. 6 Normal probability plot of residuals

Figure 5 illustrates the experimental region, with Points A, B, C, and D serving as the boundary markers defined in the previous section.

RCCD data and model fitting for IDT

The ignition delay time (IDT) for each formulation was obtained from the drop test and is presented in Table 4. To minimize systematic errors, the experiments were conducted in a randomized order, indicated by the numbers in parentheses.

Residual analysis was conducted using Minitab® to validate the assumptions of normality and independence. Figure 6 presents the normal probability plot, in which the residuals align closely with a straight line, indicating that the residuals follow a normal distribution. This plot is constructed by ordering the residuals, assigning cumulative probabilities under the assumption of normality, and plotting them on a probability scale that linearizes the normal distribution (Montgomery 2008). A good alignment with the reference line suggests that the residuals are approximately normally distributed, supporting the adequacy of the statistical model.

Figure 7 shows the plot of residuals versus fitted values, which is used to assess the independence and constant variance (homoscedasticity) of the residuals. In a well-behaved model, residuals should be randomly scattered around zero with no apparent pattern. The absence of systematic structure in this plot indicates that the residuals are independent and that the assumption of constant variance holds, further validating the regression model (Montgomery 2008).

The second-order regression model for IDT was derived using the method of least squares:

$$IDT(ms) = 24.7 - 3.67x_1 - 6.63x_2 + 8.6x_1^2 + 4.1x_2^2 + 1.25x_1x_2 \quad (4)$$

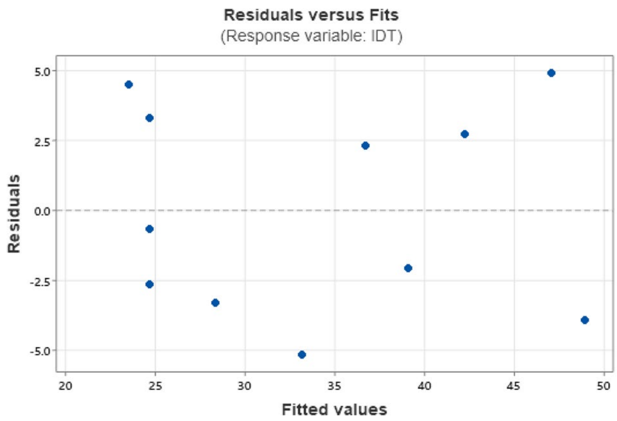

Fig. 7 Residuals versus fitted values

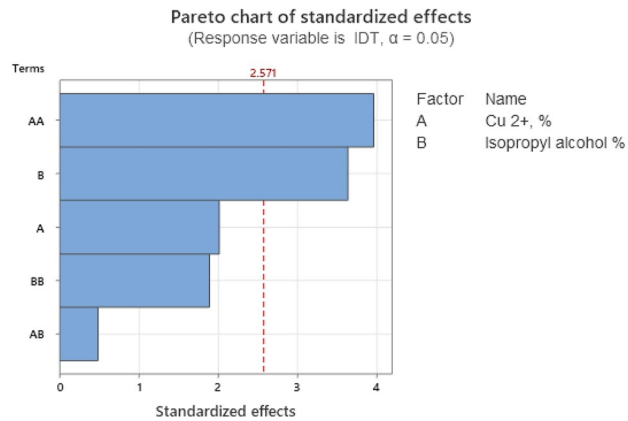

Fig. 8 Pareto chart of standardized effects

Table 5 Analysis of variance (ANOVA)

Source	df	Sum of Squares	Mean Square	F_{Calc}	F_{Tab}
Regression (R)	5	899.23	179.85	6.73	5.05
Residual (r)	5	133.68	26.74	–	–
Lack of fit (LF)	3	115.02	38.34	4.11	19.164
Pure error (PE)	2	18.67	9.33	–	–
Total	10	1032.91	–	–	–
R^2 (%)	87.06				
Adjusted R^2 (%)	85.25				

Table 6 Coded and uncoded regression coefficients

Term	Coded Coefficient	Uncoded Coefficient
Intercept	24.7	118.2
x_1 (L)	-3.67	-16.16
x_2 (L)	-6.63	-1.317
x_1^2 (Q)	8.60	0.850
x_2^2 (Q)	4.10	0.0155
x_1x_2	1.25	0.0242

where x_1 is the coded concentration of Cu^{2+}, x_2 is the coded concentration of 2-propanol, and IDT (ms) is the response variable.

ANOVA results are presented in Table 5, confirming the statistical significance of the model at a 95% confidence level.

Since $F_{Calc} = 6.73$ is greater than $F_{Tab} = 5.05$ for the regression, the model is statistically significant at the 95% confidence level. Furthermore, the lack-of-fit test resulted in $F_{Calc} = 4.11$, which is lower than the critical value $F_{Tab} = 19.164$, indicating no considerable lack-of-fit. With an R^2 value of 87.06%, these results confirm that the model adequately represents the experimental data and can be reliably used for predictive purposes.

Table 6 presents the regression coefficients in both coded and uncoded forms.

The coded coefficients provide insight into the relative importance and direction of each factor within the experimental design space. Negative linear (L) coefficients for both x_1 and x_2 indicate that increasing these factors reduces the response variable, while positive quadratic (Q) coefficients suggest the presence of curvature, indicating that an optimal region exists within the experimental range. The positive interaction term (x_1x_2) implies a synergistic effect between the two factors, where their combined influence differs from the sum of their individual effects. The uncoded coefficients can be directly applied to predict responses using actual experimental units, allowing for practical use in optimization and scale-up studies.

The Pareto chart in Figure 8 was used to assess the statistical significance of each term. Terms exceeding the reference line are significant at the 95 % confidence level.

According to Figure 8, the quadratic term for Cu^{2+} and the main effect of 2-propanol were statistically significant. In contrast, the linear term of factor A, the quadratic term of factor B, and the interaction term were not statistically significant. However, these terms were retained in the model to preserve the hierarchy structure. According to this principle, when a model includes higher-order terms, such as AA, it must also include the corresponding lower-order terms, even if they are not significant. Hierarchy ensures structural consistency and avoids the omission of potentially relevant components that may influence the effects of the higher-order terms. Furthermore, maintaining the full model helped preserve a higher coefficient of determination (R^2) compared to a reduced model without the non-significant terms (Montgomery 2008).

The response surface (Figure 9) and contour plot (Figure 10) were generated to visualize the region of minimum IDT.

Figure 9 shows the response surface plot of ignition delay time (IDT) as a function of copper (II) nitrate trihydrate concentration and isopropyl alcohol content in the formulation. A clear curvature is observed in both directions, confirming the presence of significant quadratic effects, as also indicated by the regression coefficients in Table 6. The

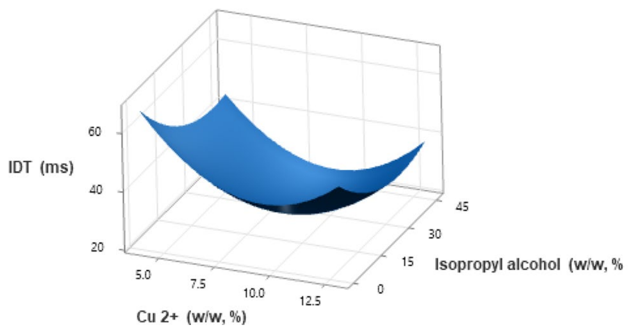

Fig. 9 Response surface for IDT as a function of uncoded Cu^{2+} and 2-propanol concentrations

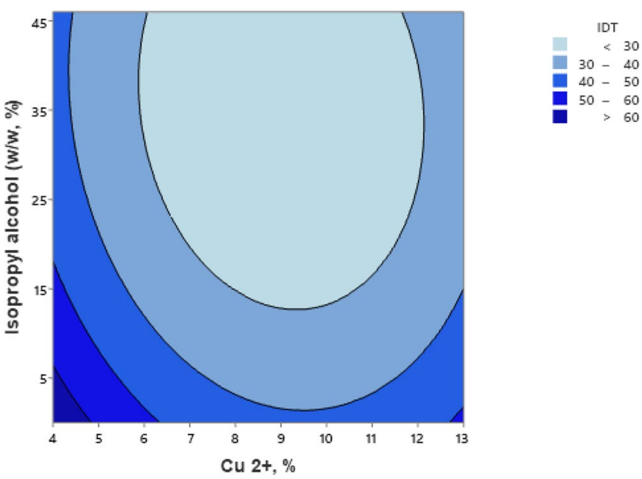

Fig. 10 Contour plot of IDT (uncoded variables) as a function of concentration of catalyst Cu 2+ (%) and concentration of isopropyl alcohol (wt%)

IDT reaches a minimum in a central region of the design space, suggesting the existence of an optimal composition for minimizing ignition delay time. At lower and higher concentrations of both Cu^{2+} and isopropyl alcohol, the IDT increases, indicating that either excess or deficiency of these components leads to slower ignition. This behavior highlights the importance of the balance between catalytic activity (Cu^{2+}) and alcohol content to promote rapid hypergolic ignition.

Fig. 10 presents the IDT lines connected to form response contours, with contour lines corresponding to IDT values of less than 30, 30–40, 40–50, 50–60, and greater than 60 seconds. These contours represent projections in the region defined by the concentrations of Cu^{2+} (%) and isopropyl alcohol (wt%), corresponding to cross-sectional profiles associated with the aforementioned IDT values.

According to Figures 9 and 10, the lowest IDT value (22 ms) was achieved at 9% copper(II) nitrate, 35.83% 2-propanol, and 55.17% MEA. In contrast, a formulation without 2-propanol yielded a minimum IDT of 42 ms. Thus, the addition of 2-propanol reduced IDT by approximately 48 %.

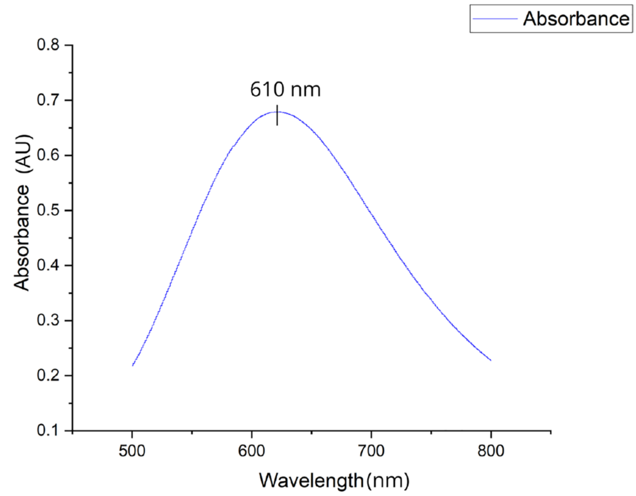

Fig. 11 Absorption spectrum (absorbance vs. wavelength) of the saturated mixture of copper(II) nitrate trihydrate and monoethanolamine

Fuel characterizaton

Complex catalyst determination

The absorbance spectrum is showed in Fig. 11.

According to Figure 11, an absorption band appears between 500 and 750 nm. The maximum peak, observed around 610 nm, corresponds to the $[Cu(MEA)_4]^{2+}$ complex (Casassas et al. 1989; Maschio 2017; Maschio et al. 2018). The presence of this complex is significant, as it catalyzes the decomposition of hydrogen peroxide and is responsible for the hypergolic ignition behavior.

NASA-CEA simulation

Figures 12 and 13 compare the specific impulse (I_{sp}) and characteristic velocity (c^*) for Formulation A (with 2-propanol) and Formulation B (without 2-propanol).

Tab. 7 shows the maximum value for I_{sp} and c^* obtained with each formulation (Formulation A with 2-propanol, Formulation B without 2-propanol, reference from Maschio et al. (2018) and Monomethylhidrazine and nitrogen tetroxide (MMH/NTO).

According to Figures 12 and 13, the formulation containing 2-propanol (35.83% 2-propanol, 55.17% MEA, and 9.0% copper (II) nitrate trihydrate) exhibited superior performance, achieving an I_{sp} of 168 s and a c^* of 1523 m/s at an O/F ratio of 4.0. In contrast, the formulation without 2-propanol reached 164 s and 1480 m/s for I_{sp} and c^*, respectively, at an optimal O/F of 3.5. Maschio et al. (2018) optimized an MEA–ethanol formulation via RCCD and reported a minimum IDT of 15.7 ms, although no thermochemical simulations were provided. Based on their reported formulation, 63.5% MEA, 27.5% ethanol, and

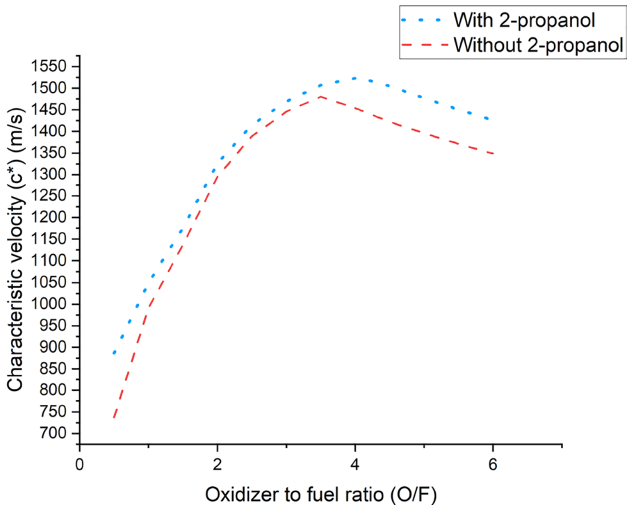

Fig. 12 I_{sp} vs. O/F curves for Formulation A (with 2-propanol) and Formulation B (without 2-propanol)

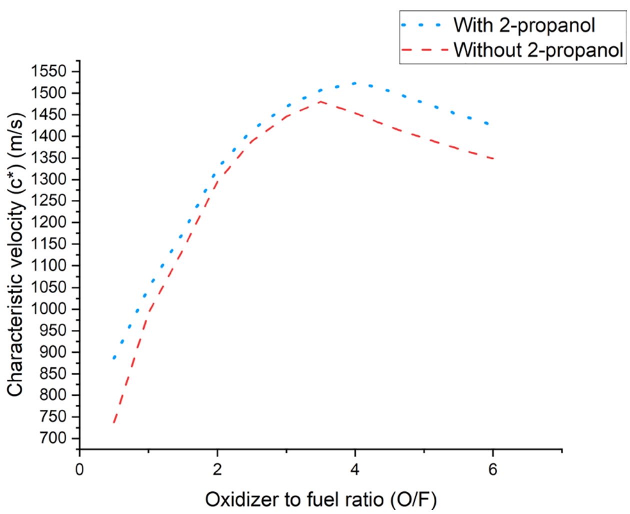

Fig. 13 c^* vs. O/F curves for Formulation A (with 2-propanol) and Formulation B (without 2-propanol)

Table 7 Optimized formulations of liquid propellant with and without 2-propanol, named as Formulation (Form.) A and B, including a reference column for Maschio et al. (2018) and Monomethylhidrazine and nitrogen tetroxide (MMH/NTO)

	Form. A (%)	Form. B (%)	Maschio et al. 2018 (%)	MMH/ NTO (%)
Monoethanolamine	55.17	91.00	63.5	(MMH fuel) 100
Copper (II) nitrate	9.00	9.00	9.0	(NTO oxidant) 100
2-propanol	35.83	-	(ethanol) 27.5	-
O/F ratio	4.0	3.5	3.5	1.6
I_{sp}, s	168	164	166	191
c^*, m/s	1523	1480	1509	1734
Comments	less toxic	less toxic	less toxic	highly toxic

Table 8 General and thermodynamic properties of fuel with 2-propanol (Formulation A) and without 2-propanol (Formulation B)

	Density (g/cm^3)	Calorific Power (J/g)	Flash Point (°C)
Formulation A	0.98155 ± 0.01	23170 ± 1	27.5 ± 0.1
Formulation B	1.08065 ± 0.01	21369 ± 1	45.5 ± 0.1

9.0% copper(II) nitrate trihydrate, we performed a simulation using NASA CEA, which yielded an I_{sp} of 166 s and a c^* of 1509 m/s at an O/F ratio of 3.5. Despite a modest increase in IDT (22 ms), the MEA–2-propanol formulation demonstrated improved energetic output and safer handling characteristics. For the reference propellant pair monomethylhydrazine and nitrogen tetroxide (MMH/NTO), the I_{sp} reached 191 s and the c^* was 1734 m/s at an O/F ratio of 1.6. However, this fuel–oxidizer combination is highly toxic to both the environment and humans. Even though the MMH/NTO pair exhibits higher performance, a greener and less toxic alternative needs to be developed to replace it. The Formulation A proposed in this study has shown potential as a viable alternative to replace more toxic propellants.

Compared with Formulation B and with the formulation proposed by Maschio et al. (2018), formulation A developed in this study demonstrates that 2-propanol–based mixtures can deliver equivalent impulse with less mass and higher exhaust velocity, due to their higher energy content per unit volume. This advantage is especially relevant for missions constrained by propellant tank volume, such as spacecraft propulsion. Therefore, the optimized 2-propanol formulation is not only more efficient but also a promising green propellant candidate. To further support its performance, experimental evaluations of density, calorific power, and flash point were conducted.

Characterization of general and thermodynamic properties

Table 8 summarizes the thermodynamic properties of the optimized formulations with (Formulation A) and without (Formulation B) 2-propanol.

The addition of 2-propanol reduced both flash point and density. These results align with its known physicochemical properties and support the improved ignition performance observed. Lower flash points correlate with shorter ignition delay times (IDTs) in hypergolic systems due to the easier formation of reactive vapors (Blevins et al. 2004). Furthermore, Formulation A's higher calorific value suggests enhanced energy release per unit mass.

Theoretical density of the fuel mixture (ρ_b) and density-specific impulse ($\rho_b I_{sp}$)

Table 9 presents the theoretical values of volumetric specific impulse ($\rho_b I_{sp}$) for the formulations with 2-propanol (Formulation A), without 2-propanol (Formulation B), and from the 2018 reference study (Formulation C) (Maschio et al. 2018). The mixture density (ρ_b) was calculated assuming the oxidizer to be 90 wt% hydrogen peroxide, with a density of 1.39 g/cm^3.

Among the tested formulations, Formulation A exhibited the highest $\rho_b I_{sp}$ value (216 s g/cm^3), slightly surpassing the other two (both 214 s g/cm^3). Although the difference may appear numerically modest, it becomes meaningful when considered alongside the lower specific mass and higher energy output of Formulation A.

This enhanced performance is primarily attributed to the thermochemical advantages of 2-propanol. Its higher heat of combustion (–2004 kJ/mol) compared to ethanol (–1367 kJ/mol) leads to greater energy release upon ignition. In addition, 2-propanol's higher boiling point (82.6 °C) and lower vapor pressure at 25 °C (45 mmHg) contribute to improved formulation stability and reduced evaporative losses (National Institute of Standards and Technology (NIST) 2024b, a). Its complete miscibility with MEA further supports the formation of homogeneous and stable mixtures.

Therefore, $\rho_b I_{sp}$ emerges as a decisive metric for evaluating the suitability of a formulation in space-constrained applications. While I_{sp} quantifies efficiency by mass, $\rho_b I_{sp}$ integrates both energetic and volumetric performance. A formulation that simultaneously delivers high I_{sp} and ρ_b, as in the case of Formulation A, provides comprehensive advantages for use in upper-stage engines, satellite thrusters, and compact propulsion systems.

Table 9 Volumetric specific impulse ($\rho_b I_{sp}$) of the formulations studied

	Specific Mass (g/cm^3)	O/F Ratio	ρ_b (mixture) (g/cm^3)	$\rho_b I_{sp}$ (s g/cm^3)
Formulation A	0.98 ± 0.01	4.0	1.28	216
Formulation B	1.08 ± 0.01	3.5	1.31	214
Formulation C	1.02 ± 0.01	3.5	1.29	214

Conclusion

In this study, a green bipropellant formulation was successfully developed and optimized, comprising monoethanolamine and 2-propanol as fuels, copper(II) nitrate trihydrate as a catalyst, and 90 wt% hydrogen peroxide as the oxidizer. Using a Rotatable Central Composite Design (RCCD), the ignition delay time (IDT) was minimized to $22 \pm 1 ms$. The inclusion of 2-propanol significantly enhanced ignition behavior, reducing the IDT by 52% compared to the alcohol-free formulation.

Beyond ignition performance, the formulation exhibited favorable thermophysical and combustion characteristics, including a density of 0.98155 g/cm^3, a calorific value of 23170 J/g, a flash point of 27.5 °C, and a theoretical density-specific impulse ($\rho_b I_{sp}$) of 216 s g/cm^3. UV-Vis spectroscopy confirmed the presence of the $[Cu(MEA)4]^{2+}$ complex, which catalyzes hydrogen peroxide decomposition and enables hypergolic ignition. NASA CEA simulations further estimated a specific impulse (Isp) of 168 s and a characteristic velocity (c^*) of 1523 m/s.

Compared to prior work by Maschio et al. (2018) using ethanol, the MEA/2-propanol formulation achieved slightly superior performance in I_{sp}, c^*, and energy density, while also improving safety via a lower flash point. Despite a modestly higher IDT, it demonstrated improved volumetric efficiency, thermal behavior, and overall system performance.

Given these advantages, the formulation emerges as a promising candidate for compact or volume-constrained propulsion systems, such as satellite attitude control thrusters or upper-stage engines. Future studies should address long-term stability, catalyst recyclability, and validation in actual engine hardware to support the practical deployment of this green bipropellant.

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Declarations

Conflicts of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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